



Within Session Exercise Sequencing During Programming for Complex Training: Historical Perspectives, Terminology, and Training Considerations

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Abstract

The primary aim of this narrative review was to outline the historical genesis of resistance training strategies that incorporate high-load, low-velocity exercises and low-load, high-velocity exercises in the same training session allowing for different “exercise sequences” to be simultaneously implemented. Discrepancies between scientific works and the terminology used within contemporary sport science publications are identified. Upon review of the literature, we propose “complex training” to be considered an umbrella term with 4 different implementations, generally used to indicate a method in which movement velocity or load is altered between sets and/or exercises within the same session with the aim of improving slow and fast force expression. We propose the following terminology for said implementations: contrast training—exercise sequence with alternating high-load and low-load (higher-velocity) exercises in a set-by-set fashion within the same session (corresponding with ‘contrast pairs’ and ‘intra-contrast rest’); descending training—several sets of high-load (e.g., back squat) exercises completed before the execution of several sets of low-load, higher-velocity (e.g., vertical jump) exercises within the same session; ascending training—several sets of low-load, higher-velocity exercises completed before several sets of high-load exercises within the same session; and French contrast training—subset of contrast training in which a series of exercises are performed in sequence within a single session: heavy compound exercise, plyometric exercise, light-to-moderate load compound exercise that maximises movement speed (i.e., external power), and a plyometric exercise (often assisted). Finally, practical applications and training considerations are presented.

1 Introduction

1.1 Background

When developing specific neuromuscular functions in team and individual sport, athletes, researchers, sport scientists, and strength and conditioning (S&C) professionals often choose to impose a range of physical training interventions [1–3]. The chronic use of heavy-load resistance training can improve maximal dynamic strength (i.e., the ability to apply external force or the ability to move similar or greater loads at higher speeds) and explosive strength (i.e., ability to increase force or torque as rapidly as possible during voluntary contractions from low or resting levels, generally

evaluated through the rate of force development) [3, 4]. Further, heavy-load resistance training can trigger positive neural (i.e., motor unit [MU] activation/recruitment and MU synchronisation), morphological (e.g., cross-sectional area and muscle pennation angle), cellular, and metabolic adaptations (e.g., fibre type composition and aerobic/anaerobic capacity) [3, 5–7]. These adaptations, in turn, lead to enhancements in athletic performance, whilst potentially reducing injury rates [8] and improving post-competition recovery [9, 10], which can promote athlete availability in training and competition.

The addition of low-load, higher-velocity exercises, such as plyometrics (movements involving fast eccentric muscle actions and muscle tendon unit stretch followed by a concentric shortening action, i.e., stretch–shortening cycle [SSC]) [11]. In a similar way, ballistic exercises (exercises where the resistance is accelerated throughout the entire movement with pure feed-forward control and without neuromuscular

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Key Points

Within a training session, the addition of light-load, high-velocity exercises (e.g., ballistic and plyometrics exercises) to high-load low-velocity exercises (e.g., heavy squats) in a complex training design does effect training outcomes relating to an athlete's lower extremity external force and velocity expression, which is founded on principles identified historically.

Specific terminology (i.e., complex, contrast, ascending, and descending training; intra-contrast rest and contrast-pairs) has been proposed and should be considered in research going forward based on the genesis of the exercise sequencing strategies.

Future research examining complex training strategies is warranted; however close attention should be given to the methodological design of the studies and the practical nature of the investigations.

feedback, wherein the time scale too short to modify movement once it has begun; e.g. isometric pull, bench throws, weightlifting derivatives) [12] both provide an effective stimulus for improving neuromuscular function in sport-specific actions [13, 14]. Thus, utilising “mixed-training” approaches that combine training exercises using heavier (i.e., strength-oriented) and lighter loads (i.e., velocity-oriented) within the same session or across multiple sessions are a rational strategy for promoting adaptations broadly within components of the force–velocity relationship compared to traditional resistance training approaches that only include a single training methodology [15–20].

The positive performance outcomes associated with these training strategies [16, 21–25] likely result from the stimulation of a broader range of adaptations within the neuromuscular system, which facilitate improved force production across a broader range of movement speeds [26, 27]. However, even with sound rationale for its implementation, determining the effectiveness of mixed-training compared to stand-alone use of a training methodology is challenging due to the lack of volume and intensity equated training studies [22]. Furthermore, it is important to consider that using loads or effort levels that evoke force production within a specific area of the force–velocity relationship could still provide enhanced performance across several loads and movement speeds (e.g., high-load, low-velocity weight training interventions can enhance high-velocity force production) [27]. Additional rationale for the implementation of mixed-training approaches includes the potential of enhancing the performance of a latter movement by performing

either light-loaded or higher-loaded movements beforehand [28, 29]. This being said, there are force–velocity demands for each exercise [19] and the load utilised with these exercises impacts the specific neuromuscular functions and velocity achieved during the performance of such exercise (and latter movements). Nonetheless, using a combination of high-load, low-velocity exercises (i.e., moving heavy loads slowly) with low-load, higher-velocity exercises (i.e., moving light or moderate loads quickly) within a single session, based on the magnitude of evidence, is likely to promote a robust upward and rightward shift in the force–velocity relationship [5, 30]. Thereby potentially enhancing the ability to express greater mechanical power [31–34] within that session and consistently throughout a training programme, which may also bring about greater expression of external force in training and competition (Fig. 1).

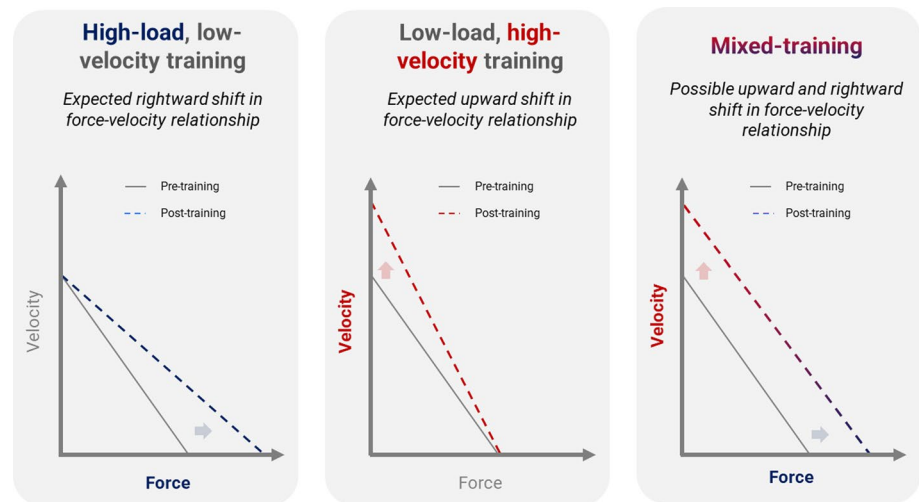
1.2 Summary of Sequences

Several researchers have investigated the within-session set sequences of high-load, lower-velocity with lower-load, higher-velocity training strategies [35–40] and there is a growing number of researchers investigating the acute responses [27, 30, 41–47] and chronic effects [14, 21, 22, 26, 27, 48–50] of these strategies. When this research is summarised, there are three within-session sequences that are commonly investigated:

- A- Several sets of high-load (e.g., back squat) exercises completed before the execution of several sets of low-load, higher-velocity (e.g., vertical jump) exercises within the same session.
- B- Several sets of low-load, higher-velocity exercises completed before several sets of high-load exercises within the same session.
- C- Alternating high-load and low-load (higher-velocity) exercises in a set-by-set fashion within the same session.

However, discrepancies exist in the terminology used to name these training sequences as outlined in Table 1. For example, Cormier et al. [26] defined sequence C as “complex training” (CPX) and sequence A as “contrast training” (CNT) based on definitions by Ebben et al. [51]. In contrast, Marshall et al. [27] used the definitions outlined by Duthie et al. [52] that defined sequence A as CPX, sequence B as “traditional training”, and sequence C as CNT. In brief, what the former study [26] classified as CPX training, the latter [27] considered to be CNT, and vice-versa. These divergent definitions are problematic as they reflect discrepancies observed in the wider literature and other meta-analyses (Table 1) on the topic, which have also used these definitions interchangeably [21, 49, 50]. Thus, it is important to clarify

Fig. 1 The intended changes in the force-velocity relationship following the use of high- or low-load versus mixed-training



the terminology regarding these training methods utilising different exercise sequences.

When summarising evidence relating to the adaptations evoked by these exercise training sequences, meta-analyses often report a wide range of adaptations [21, 26, 27, 48–50]. While assessing magnitude of change to determine the efficacy of these training interventions, researchers typically use effect size (ES) scales, which allow results from different studies to be directly compared (e.g., scale thresholds: <0.2 [trivial effect], 0.2–0.49 [small effect], 0.5–0.79 [moderate effect], >0.8 [large effect]) [53] or specific scales for the determining the magnitude of training effects in S&C [54]. Although, these ES scales are reported with the purpose of applying and comparing findings between studies, non-uniform language and implementation regarding participant characteristics (e.g., performance calibre and initial strength levels) and intervention characteristics (e.g., training volume, exercise intensity, rest intervals, and training duration) pose a problem. Therefore, it is important for research concerning CPX to address these issues and provide clear classifications. Thus, when possible, in this narrative review, data are described in reference to strength status of individuals [17, 55, 56] and performance calibre classification [57].

1.3 Underpinning Mechanisms

The adaptations leading to the performance gains evoked by mixed-methods training approaches are thought to be induced by the acute (within-session) responses to an early exercise set. However, when inspecting chronic changes over time, these training approaches also allow for training variation (e.g., manipulation of set recovery and exercise intensity), training efficiency, and ability to perform strength training sessions whilst avoiding sustained high-volume training. This strategy, acutely, facilitates enhanced performance in later exercises (potentiation) through a

conditioning activity (CA), a prior excitation of skeletal muscle, commonly a high-load resistance training exercise [43, 58], which allows for a greater overall adaptation to be evoked. This can be accomplished by grouping exercises of similar load and velocity at the beginning of the session and performing exercises of lesser or greater load and velocity at the end of the session (e.g., all heavy before light), or by alternating these exercises to elicit this muscle excitation [59]. Thus, greater overall loads are lifted, or movement speeds are attained, in the subsequent exercise sets or tasks within the session, and a greater adaptive stimulus is imposed [35, 60, 61]. For example, acute physiological, biomechanical or psychological responses to a set of intense back squat exercise performed using relatively high loads (relatively slow speed) might enhance force and/or velocity in a subsequent lower-load, higher-velocity exercise (e.g., sprint or jump), if sufficient rest is allowed between exercises [30]. However, it may be important to consider that in cases where only a few repetitions are performed within a CA, and hence fatigue is minimal, a brief recovery period may still be necessary to reduce motor pattern interference effects [62], which may reduce performance in the subsequent exercise (since force–velocity characteristics differ) [63].

The adaptations described above are often referred to as a “post-activation potentiation” (PAP) effect; however, it must be noted that PAP is classically defined as an enhancement in muscle force resulting predominately from phosphorylation of the myosin light chain in type II fibres and usually verified in humans by electrically evoked muscle twitch [28–30]. More specifically, with PAP, the CA stimulates an influx of sarcoplasmic calcium (Ca^{2+}) into the myoplasm that activates myosin light chain kinase, which phosphorylates the light chains and thereby promotes actin-myosin cross-bridge cycling due to greater Ca^{2+} sensitivity, consequently leading to improvements in submaximal force production [30, 64–66]. Yet, this enhancement has a short

Table 1 Summary of intervention outcomes from six meta-analyses on complex training with sequence and terminology implemented

Meta – Analyses	Sequence (terminology used)	Analysis	1 – RM squat	Sprint time	Vertical jump height	Change of direction time	Peak power	Peak velocity	Peak force
*Bauer et al. [21]	<i>C (complex)</i>	<i>vs control</i>	23.6% [16.4, 30.7], $p < 0.001$	5 m: –7.8% [–14.8, –0.9], $p = 0.034$, 10 m: –4.1% [–6.0, –2.1], $p = 0.001$, 20 m: –4.4% [–7.4, –1.4], $p = 0.012$, 30 m: –4.3% [–8.0, –0.6], $p = 0.032$, 40 m: –4.2 [–8.9, 0.5], $p = 0.068$	CMI: 8.9% [5.6, 12.3], $p < 0.001$; SJ: 12.7% [8.0, 17.4], $p = < 0.001$				
				<i>vs alternative</i>	7.0% [0.2, 13.7], $p = 0.044$				
				5 m: –0.7% [–2.3, 1.0], $p = 0.323$, 10 m: –0.8% [–2.4, 0.9], $p = 0.325$, 20 m: –0.9% [–1.6, –0.1], $p = 0.028$, 30 m: –1.1% [–4.9, 2.8], $p = 0.357$, 40 m: –2.3 [–4.7, 0.2], $p = 0.061$					
				10 – 40 m: –0.69 [–1.02, –0.36], $p < 0.001$	CMI: 0.85 [0.45, 1.25], $p < 0.001$	–0.83 [–1.08, –0.59], $p < 0.001$			
				10 – 40 m: –0.94 [–1.33, –0.54], $p < 0.001$	CMI: 0.88 [0.42, 1.34], $p < 0.001$	–1.17 [–1.43, –0.90], $p < 0.001$			
				10 – 40 m: –0.51 [–0.90, –0.11], $p = 0.01$	CMI: 0.50 [0.19, 0.81], $p = 0.002$				
				10 – 40 m: –0.27 [–0.92, 0.39], $p = 0.43$	CMI: 0.55 [0.29, 0.81], $p < 0.001$	–0.68 [–1.20, –0.15], $p = 0.01$			
Cormier et al. [26]	<i>C (complex)</i>	<i>vs control</i>	1.61 [1.12, 2.10], $p < 0.001$						
		<i>Post-pre</i>	2.01 [1.18, 2.84], $p < 0.001$						
		<i>vs control</i>	1.38 [0.30, 2.46], $p = 0.01$						
		<i>Post-pre</i>	1.29 [0.61, 1.98], $p < 0.001$						

Table 1 (continued)

Meta – Analyses	Sequence (terminology used)	Analysis	1 – RM squat	Sprint time	Vertical jump height	Change of direction time	Peak power	Peak velocity	Peak force
Freitas et al. [48]	<i>C (complex training)</i>	<i>vs control</i>		15 – 30 m: –0.83 [–1.65, –0.01], $p=0.05$	CMJ: 0.62 [0.11, 1.13], $p=0.02$				
		<i>Post–pre</i>		15 – 30 m: –0.74 [–1.22, –0.25], $p=0.003$	CMJ: 0.27 [–0.02, 0.57], $p=0.07$				
Marshall et al. [27]	<i>A (complex)</i>	<i>Post–pre</i>	0.93 [0.25, 1.60], $p=0.007$		SJ & CMJ: 0.61 [–0.08, 1.31], $p=0.08$				0.14 [–0.50, 0.78], $p=0.67$
	<i>B (traditional)</i>		1.19 [0.76, 1.63], $p<0.001$		SJ & CMJ: 0.41 [0.12, 0.70], $p=0.006$		0.52 [0.24, 0.80], $p=0.0003$	0.24 [–0.28, 0.76], $p=0.36$	0.14 [–0.24, 0.52], $p=0.48$
	<i>C (contrast)</i>		1.16 [0.70, 1.62], $p<0.001$		SJ & CMJ: 1.10 [0.40, 1.80], $p=0.002$		0.67 [0.20, 1.13], $p=0.005$	1.01 [0.40, 1.62], $p=0.001$	0.64 [0.06, 1.23], $p=0.03$
Pagaduan et al. [49]	<i>C (contrast)</i>	<i>vs control</i>			CMJ: 1.46 [0.46, 2.46], $p=0.004$				
		<i>vs alternative</i>			CMJ: 1.30 [0.31, 2.30], $p=0.01$				
Thapa et al. [50]	<i>C (complex)</i>	<i>vs control</i>		5 m: 1.91 [0.81, 3.0], $p<0.001$, 10 m: 0.92 [0.32, 1.53], $p=0.003$, 15 m: 1.07 [0.31, 1.83], $p=0.006$, 20 m: 1.04 [0.49, 1.58], $p<0.001$, 40 m: 1.28 [0.80, 1.76], $p<0.001$					

Data are presented as overall ES [95% CI] with reported p value. *Bauer et al. did not report ES but did report overall percent change

I-RM one repetition maximum, *CMJ* countermovement jump, *SJ* squat jump, *Change of direction* consisted of skilled-based and traditional tests (e.g., t-test and 505 agility), *Peak power* highest power recorded during concentric portion of movement, *Peak velocity* highest velocity recorded during concentric portion of movement, *Peak force* highest force recorded during concentric portion of movement

Alternative is classified as any other stand-alone method (e.g., traditional resistance training)

half-life of ~28 s [29], whereas the improvements in force production following CAs often occur between 6 and 10 min [29, 43, 47]. Therefore, when changes in voluntary force or overall function are observed, the term “post-activation performance enhancement” (PAPE) has recently been proposed [28, 29, 67–69], although this terminology has yet to be fully agreed upon. The alternative mechanisms proposed for PAPE include increases in muscle temperature, muscle/cellular water content, and muscle activation, or alterations in motor pattern which is verified via a voluntary task (see Fig. 2.) [29]. Moreover, although these mechanisms are influential depending on the characteristics of the training programme (e.g., warm-up, recovery periods, and CA intensity), currently, we do not know which acute changes truly underpin the acute performance enhancements achieved. Therefore, whilst contemplating the rationale for implementing mixed-training approaches or conceptualising a training programme, it is important to keep these alternative mechanisms and their potential effects in mind (see [28, 29, 67] for reviews).

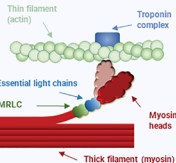
1.4 Purpose

Since the introduction of these CPX strategies, many acute response and intervention studies have been completed (Fig. 3). As such, the aforementioned inconsistencies in the terminology used in these studies pose a problem in the general sport science and rehabilitation community, as many of the identified studies included in these reviews have adopted different terminologies to describe, or define, the same interventions. Consequently, an up-to-date review is needed to provide clarity in relation to the definitions of such terms. This would enable clarity in study design as well as the subsequent interpretation and aggregation of results. Accordingly, the aims of this paper are to: (1) provide a historical perspective, including identification of the timeline and

earliest appearance of these exercise sequences in the literature; (2) clarify definitions for practitioners and researchers whilst discussing the potential the underlying mechanisms that determine their effectiveness; and (3) make programming recommendations for the use of these sequences for practitioners.

To identify the timeline of the original publications the following strategy was used. First, systematic reviews with meta-analyses published between 2019 and 2021 on exercise sequence groupings and sports performance outcomes were assessed. These reviews [21, 26, 27, 48–50] followed the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), and only included peer-reviewed literature in their analysis. They all individually used appropriate tools to limit the risk of bias of the studies included in their reviews thereby assembling only papers of “good” quality as defined by their risk of bias assessment. Altogether, their inclusion/exclusion criteria resulted in the inclusion of intervention studies (≥ 3 -week periods with many accrued sessions), with recreationally active to highly trained calibre participants [57]. All papers (authors, titles) were imported into an Excel™ (Microsoft Corporation, Redmond, WA, USA) form, with all duplicates removed, and full-text papers sourced. Two reviewers (PC and TTF) carefully read each paper to extract the definition used in the article, the exercise sequence performed, and the cited reference (s) from which the definitions were obtained (corresponding in-text citation with reference list). After the relevant references were searched, the most frequently used definitions at their earliest chronological appearance in the literature were extracted to form a timeline of publication dates and to identify the first definitions reported in scientific works. This procedure provided the authors of the present paper a better understanding of the genesis of these methods. The full texts of the respective peer-reviewed journal articles were sourced to identify the actual definitions reported.

Fig. 2 Depiction of the potential underpinning mechanisms to the performance enhancements following complex training. Image has been adapted from Blazeovich & Babault [29]

	PAP	PAPE
	POST-ACTIVATION POTENTIATION	POST-ACTIVATION PERFORMANCE ENHANCEMENT
Performance effects	 Enhancements in muscle twitch properties	 Enhancements in voluntary dynamic force production
Potential time course	 Immediate, transient effects Starts within seconds following conditioning activity and has a short half life (~28 seconds)	 Delayed, long-lasting effects Starts minutes following conditioning activity and lasts 6–10 minutes
Potential mechanism(s)	 Myosin regulatory light chain (MRLC) phosphorylation MRLC phosphorylation and subsequent rotation of the myosin head (light color) increases the probability of the head attaching to actin, and thus force production	 Muscle temperature  Muscle & muscle fiber water content  Muscle activation (including motivation) • muscle temperature are associated with • rate of force development and shortening velocity in both fast- and slow-twitch muscles/fibers. • blood flow & intracellular (myocellular) water can enhance muscle force production, and this effect is greater in type II fibers. • level of voluntary neural drive from conditioning contractions may improve muscle activation, which can • maximum voluntary rate of force development and maximal muscle force.
Inhibition	 Fatigue & motor pattern interference	 Fatigue & motor pattern interference

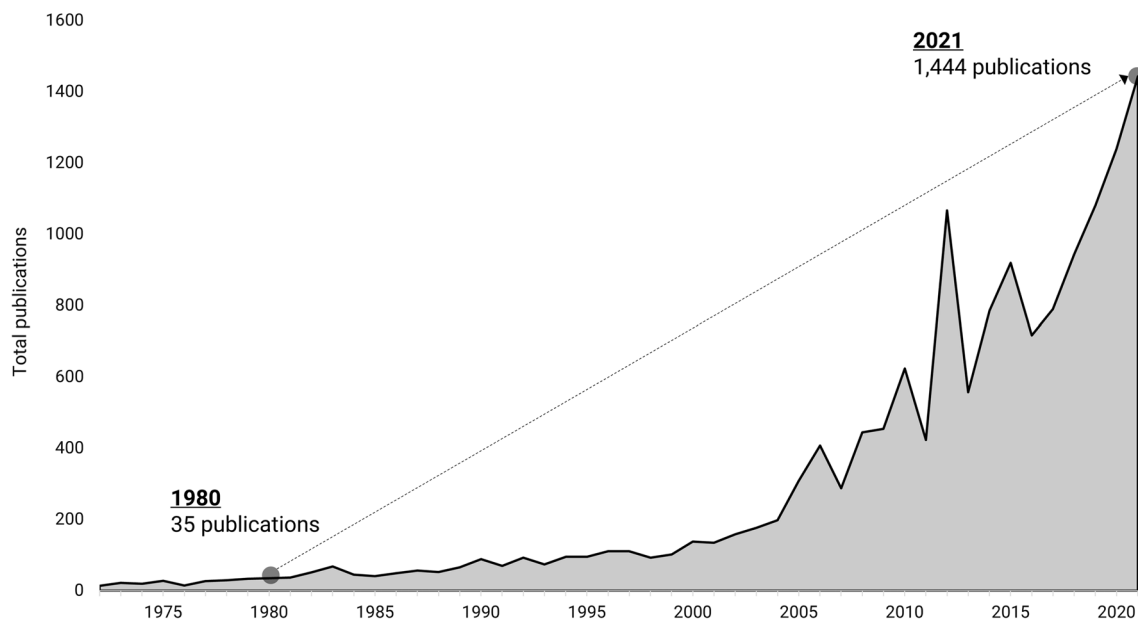


Fig. 3 Yearly publications (1972–2021) based on the following Boolean search phrase: “complex training” OR “contrast training” OR “exercise sequence” OR “French contrast training” OR “post activation

potentiation” OR “post activation performance enhancement”. Data extracted from <https://app.dimensions.ai/discover/publication>

Finally, the authors electronically and manually searched for other sport science books/articles to outline additional empirical, anecdotal, and historical evidence.

2 Historical Perspective

In Fig. 4, the genesis of these methods from early literature are presented. The differing definitions and citations used has led to confusion amongst researchers. Accordingly, this confusion, and subsequent discrepancies in terminology, are reflected in a large number of acute studies (i.e., time-course of potentiation in one session following conditioning contractions) and reviews on the topic [27, 41–47, 70–73], hence the importance of clarity.

The first definitions of training interventions utilising both high-load and lower-load exercises of various sequences within a session are difficult to identify. To the authors’ knowledge, these methods were first explicitly described in publication by Russian sport scientists Verkhoshansky and Tatyana in 1973 [74]. Although these methods were almost certainly used in practice prior to this publication, the specific use of this terminology does not seem to have been previously documented. In one of his publications [75], Verkhoshansky speaks of a researcher named Lelikov who experimented with “dynamic resistance exercises with different movement rates...” with first mention of the CPX method. Further explained is the rationale: “the complex method utilizes the body capacity to summarize the adaptive

reactions of different training stimuli in a resulting training effect, which is higher than the sum of the training effects of each exercise”. His colleagues’ experiments on CPX methods are also described in his book, where the “stimulation method” consisting of the combination of barbell squats (90% 1-RM [one repetition maximum]) with barbell vertical jumps (50% 1-RM) was compared to the stand-alone use [75]. This “mixed-training” approach resulted in a greater increase in unloaded vertical jump power output (i.e., 90% 1-RM squat: ~8%, 50% 1-RM VJ: ~16%, and combined: ~20%). Furthermore, also provided are definitions on CNT for the development of high-speed strength as a form of the CPX method. In Verkhoshansky’s book, it is written: “The goal is to achieve a contrast of the kinesthetic sensations, performing, at the maximal power output, a comparable movement in alternating conditions, more or less difficult in comparison to the normal”. Another example extracted from his book of CNT use with track and field athletes, focuses on a heavy resisted sprints immediately followed by a towed sprint (at 2–4% of athletes’ body mass).

In 1986, Fleck and Kontor [76] published notes from a study tour that was undertaken at the Moscow Institute of Sport with a cohort of US and Canadian S&C coaches. The authors described Verkhoshansky’s definition of CPX as “a series of several exercises performed in succession with the goal of the entire movement complex [for] the improvement of one physical characteristic” to be implemented in the special preparation phase of training of the periodised plan. In 1987, Roll and Omar [77] also wrote about the use of

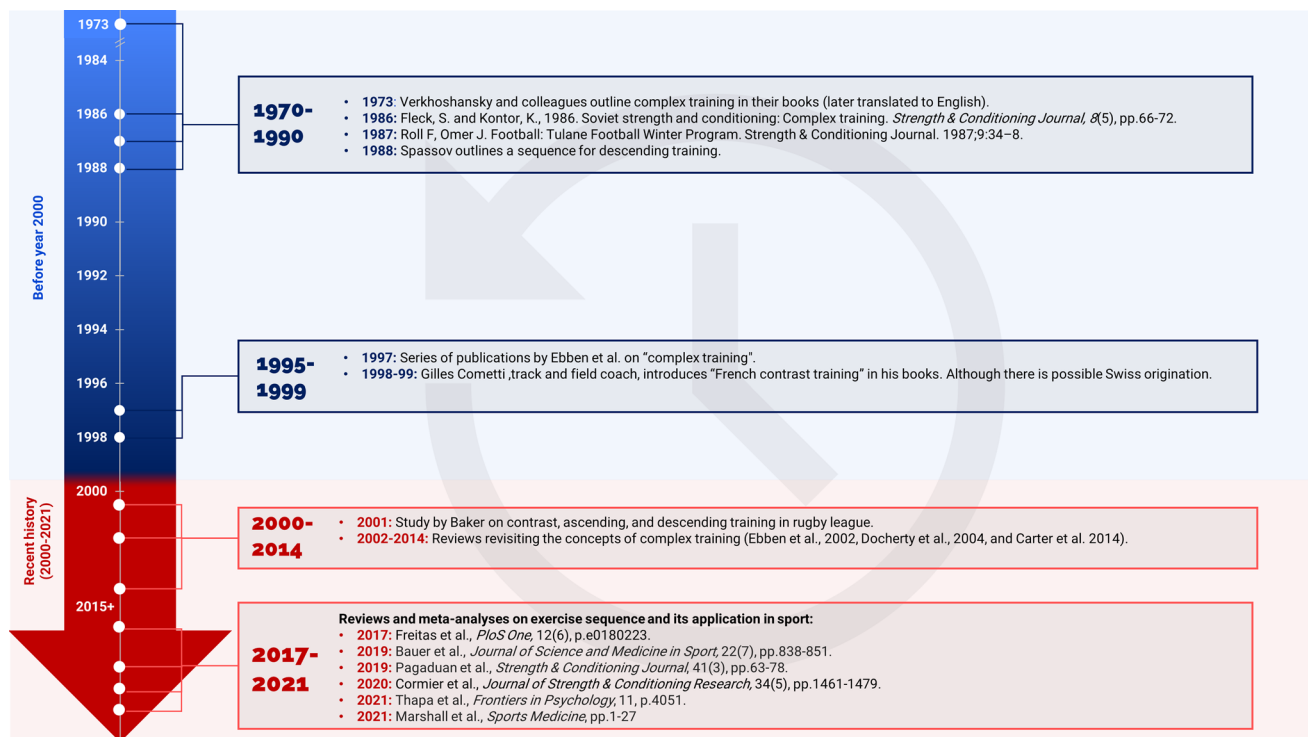


Fig. 4 Timeline from 1970 to 2021 to better understand the historical perspectives

"complexes" in their periodisation scheme with American football players where "*contractions against heavy resistances and slow movement velocities (i.e., heavy squats, cluster benches) followed by exercises using lighter resistances of fast velocities (i.e., plyometrics)*" are used with the purpose of increasing power expression by incorporating the higher velocity movements into their training programme (strength/power phase). It is likely that these articles are the basis upon which Ebben et al. [51, 78, 79], described CPX as a workout session that "*alternates biomechanically comparable high-load weight training and plyometric exercises in the same workout*", which has been adopted widely in contemporary literature. Furthermore, this is why terms such as strength-power potentiation "complexes" or "complex pairs" [76, 78, 80] have been introduced to define the potentiated pairings aligned with CPX.

In the 1990s to early 2000s, Cometti popularised a method of sequencing exercises named "French" CNT (FCNT) [81, 82]. Dietz and Peterson [82] have further popularised this method with empirical and anecdotal evidence in books and online content. The set structure follows this sequence: a heavy compound exercise, a short SSC plyometric exercise, a light-to-moderate load compound exercise that maximises external power production (i.e. movement speed), and a plyometric exercise (often assisted) [83, 84]. The aims are to elicit a PAPE with the contrasted nature of loading/contraction types whilst also targeting four

components on the force-velocity relationship (i.e., in the specific order: maximum strength, speed-strength, strength-speed, and maximum speed, respectively) [83, 84]. Furthermore, this method is a subset of CNT, which is used by many practitioners in the field [2, 82, 85]; however, only recently has evidence been produced relating to its effectiveness in acute ($11 \pm 6.3\%$ increase in countermovement jump [CMJ] height post-pre FCNT) [83] and longitudinal training studies ($+3.8\%$ Sargent jump, 1.8% in CMJ height, and 2.3% in seated medicine ball throw distance in 8 weeks) [84]. Hence, there is limited empirical evidence in Western literature supporting this specific exercise grouping.

When considering the sequence in which several sets of high-load exercises are completed before the execution of several sets with light- or moderate-loads moved at higher velocities, or the opposite sequence, there is a variety of terms used in the literature. However, these methods are described as a form of CPX by Spassov [86] and Baker [59]. It is termed the descending training (DT) whereby the high-load exercises are completed first, working towards the lower loads performed later in the workout session (i.e., Bulgarian method) [59, 86]. Alternatively, ascending training (AT) requires that lighter loads are lifted before heavier loads within a workout session [59, 87]. Years later (2001), Baker [59] conducted a study in rugby league athletes to compare the effectiveness of AT, DT, and CNT. The rationale for the implementation of CNT from Baker [59] is as follows:

“by alternating the heavy and light load from set to set, the neural system is ‘super stimulated’, making the lighter power load ‘feel lighter’”. Hence, improving the mechanical power expression in the lighter exercise set.

3 Proposed Terminology

When examining the historical genesis of these methods, it becomes evident that many researchers have inconsistently interpreted this terminology, despite it being clearly defined in these early works, which is a consistent problem in this field [88–90]. Due to these misinterpretations, there are discrepancies in how these methods are defined in the contemporary literature, which become problematic when comparing findings. Thus, we propose potentially clearer terminology in Fig. 5 based on the early definitions and on the historical perspective described above. Note that since CPX training is proposed as an umbrella term with four possible implementations, the terminology can be combined with the implementations when describing the training methods (e.g., CPX-CNT, CPX-DT, CPX-AT, and CPX-FCNT). To align with the terminology outlined, and somewhat contrary to the existing literature, we propose the use of ‘contrast-pairs’ rather than ‘complex-pairs’, and for the recovery periods, ‘intra-contrast rest’ periods preferred

to ‘intra-complex rest’, since they relate specifically to the pairings and rest associated with CNT and FCNT. For this narrative review, this proposed terminology will be used throughout these coming sections.

4 Training Considerations and Practical Application

As outlined in the following sections, CPX is heavily influenced by several programming characteristics such as intervention duration and frequency (sessions per week), recovery intervals (intra-contrast and inter-set), CA intensity, sets/repetitions; and participant characteristics such as athlete strength levels, muscle fibre composition, muscle–tendon properties, and athlete performance calibre (e.g., recreationally trained vs elite), which must be taken into consideration when programming.

4.1 Duration and Frequency

Based on recent meta-analyses, CPX intervention lengths of a minimum of 3 weeks are required to evoke adaptations across a range of different physical performance parameters [26, 27, 48]. However, training durations of 6 weeks [26, 48] and 8 weeks [50] have resulted in positive adaptations

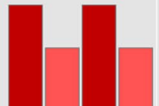
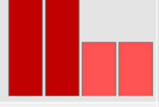
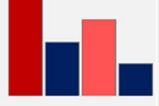
Training terminology 	Training description 	Exercise intensity 	Recovery interval 	Examples
Complex	Umbrella term with 4 different implementations, generally used to indicate a method in which movement velocity or load is altered between sets and/or exercises within the same session with the aim of improving slow and fast force expression.			
Contrast 	Exercise sequence with alternating high-load and low-load (higher-velocity) exercises in a set-by-set fashion within the same session (corresponding with ‘contrast pairs’ and ‘intra-contrast rest’)	• Conditioning activity (CA): 0–85% 1-RM • Subsequent task: BM to 60% 1-RM	Intra-contrast rest: Strong/trained athletes: 5–7 min (experiment to individualize); Weaker/recreationally trained: ≥ 8 min Inter-set rest: 3–4 min	1. Back squat 85% 1-RM 2. CMJ 3. Back squat 85% 1-RM 4. CMJ
Ascending 	Several sets of low-load, higher-velocity exercises completed before several sets of high-load exercises within the same session	• Light-load: BM to 60% 1-RM • Heavy-load: >85% 1-RM	3–4 min between sets	1. CMJ 2. CMJ 3. Back squat 85% 1-RM 4. Back squat 85% 1-RM
Descending 	Several sets of high-load (e.g., back squat) exercises completed before the execution of several sets of low-load, higher-velocity (e.g., vertical jump) exercises within the same session	• Heavy-load: >85% 1-RM • Light-load: BM to 60% 1-RM	3–4 min between sets	1. Back squat 85% 1-RM 2. Back squat 85% 1-RM 3. CMJ 4. CMJ
French contrast 	Subset of contrast training in which a series of exercises are performed in sequence within a single session: heavy compound exercise, plyometric exercise, light-to-moderate load compound exercise that maximizes movement speed (i.e., external power), and a plyometric exercise (often assisted).	• Heavy compound: 80–90% 1-RM • Plyometric: BM • Light-to-moderate compound: 40% • Plyometric: assisted	Intra-contrast rest: 20s between exercises; and 4–5 min rest between each series performed	1. Back squat 85% 1-RM 2. CMJ 3. Jump Squat 30% BM 4. Band Assisted CMJ

Fig. 5 Proposed terminology and summary of training recommendations for complex training. *BM* body mass, *1-RM* one repetition-maximum, *CMJ* countermovement jump

in both COD speed and jump heights with trained/highly trained participants. The optimal training frequency for improvements in sprint and jump performances was reported to be 2–3 sessions per week (for a total of 12–18 sessions) in 6-week interventions with team-sport athletes within the same performance calibre based on recent meta-analyses [26, 48]. Some studies have reported improvements of 12 weeks duration (2 sessions per week CNT) [91]. However, there are few longitudinal experiments with long study durations (> 12 weeks) to identify the point at which diminishing return may present. Nonetheless, these recommendations may differ according to the athlete's training experience or strength level, and the characteristics of the training programme implemented [43]. In the only investigation [92] comparing the effectiveness of CNT performed once versus twice a week (vs control) for 6 weeks, no significant differences were observed between conditions, with improved sprint times (−9.2% and −6.2%, respectively) and increased jump heights (12.6% and 9.6%, respectively) with highly trained/national level soccer players. However, higher frequencies including 4 sessions per week (for 5 weeks in highly trained/national rugby athletes), have also triggered improvements in upper and lower-body strength (4–9% in 1-RM bench press and 9–12% in the back squat) and sprint running times (−1 to −2% in 10 to 30-m runs) [93]. Similarly, 4-sessions/week frequencies have led to improvements in vertical jump height (~2.5–8.5%) in already jump-trained athletes (i.e., volleyball) [94, 95]. Based on the aforementioned research, a minimum of 3 weeks is recommended to elicit favourable adaptations in various physical performance tasks with frequencies of 2–4 sessions per week. Little research has examined the effects of CPX frequency on performance outcomes, or directly comparing frequencies between these methods and traditional strength training; thus, further research is required before strong recommendations can be made.

4.2 Recovery and Intensity

Due to the nature of the set structure used in CNT, specific terminology is required to describe the recovery periods. The *intra-contrast rest* is the recovery interval between the CA and subsequent exercise (contrast pairs), and the *inter-set rest* is the recovery period between contrast pairs. Concerning intra-contrast rest, Seitz and Haff [42] reported that stronger individuals see greater effects (ES) with 5- to 7-min intervals (0.62) whereas weaker individuals exhibit greater effects with ≥ 8 min intra-contrast rest intervals (0.36) in both lower and upper body exercises. Furthermore, the authors indicated that the CA type and intensity may influence the post-CA enhancement. For example, plyometric CAs elicited greater ESs than moderate (30–84% 1-RM), heavy resistance (≥ 85% 1-RM), and maximal isometric

(0.47 vs 0.41, 0.19, and −0.09, respectively) CAs on subsequent jumping, sprinting, throwing, and upper body ballistic task performance. Similarly, Wilson et al. [43] reported greater ESs in trained than recreationally trained male and female individuals (0.81 vs 0.14, respectively) with significant differences in effect between moderate (60–84% 1-RM) and heavy (≥ 85% 1-RM) CA intensities (1.06 vs 0.31). Maloney et al. [47] reviewed the effect of heavy resistance training (with loads > 85% 1-RM) and ballistic exercises as a CA for PAPE inducement also with recreationally trained and trained males and females and found similar acute post-CA improvements using both CA intensities. However, they also reported that short recovery times between ~30 s and 3 min using maximal, lower-load exercises as the CA (e.g., depth jumps and weighted jumps) resulted in the best post-CA enhancement. This is consistent with the higher ESs found by Seitz and Haff [42] using plyometrics and moderate load CAs. Accordingly, individual athletes might benefit most from specific recovery intervals and CA intensities to elicit PAPE. Interestingly, light-load CAs appear to elicit greater post-CA performance enhancements in strength-trained individuals. This may suggest that lighter (and accordingly faster) CAs may be superior in this instance. It could be that performing “slow” CAs increases muscle temperature whilst also imposing a motor pattern interference, whereas the light-load CA (fast) movements better resemble (less motor pattern difference) the subsequent lighter-load task. Hence, there could potentially be less motor pattern interference [29, 63].

Furthermore, the number of sets and repetitions are also an important consideration. Wilson et al. [43] recommend the use of multiple CA sets versus single sets to induce post-CA performance enhancements. There is limited research on the optimal number of sets of CNT pairings; however, it appears that the lower-body PAPE effect can be observed over a range of at least 1–3 sets performed at moderate to high loads (60–100% 1-RM) with trained subjects, although even this is highly individual [96].

To provoke classical PAP, two dilemmas outlined by Sale [30] must be considered. First, prolonged/intense CAs may lead to greater PAP but also greater levels of fatigue. Second, the longer the intra-contrast rest periods between the CA and the subsequent task, the greater the recovery from fatigue, and (the time sensitive) motor pattern interference, but also the greater the attenuation of the PAP mechanism. However, as previously mentioned, PAPE is evoked with a greater time lag than PAP. Which could mean that muscle temperature, muscle water, and muscle activation (and potentially others) may be the main mechanisms underpinning the performance enhancements in voluntary force, which could explain the time lag (5–7 min in stronger individuals) commonly observed [42, 55]. Nonetheless, the same concept outlined by Sale [30] applies; that some optimum time between

performance and enhancement effects and reducing fatigue must exist. Another dilemma that is rarely discussed when targeting the PAPE mechanism within CNT, is that the inter-set rest period between contrast pairs is not well-reported in intervention studies. Consequently, large variations in inter-set rest periods have been reported (2–10 min) [38, 76]. Furthermore, potentiation, as previously mentioned, is highly individual, S&C professionals are encouraged to experiment with exercises and rest periods with individual athletes before implementing them in long-term interventions. Based on this research, 5–7 min intra-contrast rest periods are suggested for stronger individuals (relative strength ≥ 1.75 kg/kg [1-RM in kg divided by body mass BM] in kg) in the back squat, ≥ 1.35 kg/kg in the bench press for men and women, respectively), and ≥ 8 min for weaker individuals to optimise CA-induced performance with CAs consisting of ballistic exercises (e.g., depth jumps, weighted jumps, or weightlifting and derivatives), plyometric exercises (e.g., continuous jumps), or moderate-load exercises (e.g., squats at 60–85% 1-RM). Currently, the effects of inter-set rest on long-term adaptations remain largely unknown. Furthermore, most literature reports the time-course to peak performance [55] enhancement in voluntary tasks wherein the peak performance appears to be highly individual in nature; thus, researchers and practitioners should consider this whilst implementing the aforementioned recommendations and accordingly with future research in this area.

4.3 Strength Characteristics

The physical characteristics of individuals are important moderators in the potentiation observed acutely and improvements detected over longitudinal interventions. Stronger individuals (relative strength ~ 2.0 kg/kg in the back squat 1-RM) exhibit greater potentiation acutely than weaker individuals (~ 1.75 kg/kg ratio), with a meaningful magnitude of potentiation occurring earlier in the stronger subjects using ballistic and non-ballistic CAs [55, 56]. Contrarily, Seitz and Haff [42] found that subjects with relatively greater strength levels (relative strength ≥ 1.75 kg/kg) exhibited only slightly larger ES post-CA jump enhancement (stronger: 0.41 vs weaker: 0.32). Thus, perhaps, relative strength level thresholds of ~ 2.0 might be preferable to 1.75 in the back squat for a meaningful effect to occur acutely. Furthermore, participants with more than 2 years' resistance training experience exhibited larger ESs compared to recreationally active individuals or those with less than 2 years' resistance training experience (0.52, 0.44, and 0.07, respectively). As presented in previous literature, recreationally active 'untrained' athletes initially achieve greater adaptations in the early phase of a programme, and stronger athletes see marginal gains in strength and power output [7, 17]; however, stronger subjects tend to elicit greater

post-CA enhancement [56]. Still, perhaps when using CPX methods, the training provides a significant overload stimulus, which untrained/weaker athletes do not tolerate well. Yet, trained/stronger athletes can better cope with training loads and overall training volume [97], and may require, a varied training stimuli; thus, CPX could be inferred as an advanced technique that may be appropriate after the athlete has developed the pre-requisite training history and levels of strength. Based on the available evidence [42, 70, 72], to prescribe CNT, the athletes should have adequately familiarised with resistance training, and a pre-requisite level of strength (kg/kg) of ~ 2.0 in the back squat exercise, and ≥ 1.35 in the bench press regardless of sex [42]. However, perhaps the loads, volume and rest periods will differ, and the largest acute effects will ultimately be seen in those with greater strength levels and also dependent on the exercise performed. Whether this then translates into a greater long-term adaptation, especially since less adaptive potential resides in already-trained individuals, is not yet clear.

4.4 Performance Calibre

The performance calibre of the athlete could also be a factor to consider when prescribing CPX since the post-CA enhancement that practitioners are attempting to target is highly variable [98]. Cormier et al. [26] observed that when trained and highly trained (semi-professional and professional, relative strength ~ 1.6 – 1.7 kg/kg in the back squat/half-squat) team-sport athletes were compared to recreationally trained amateur athletes (relative strength ~ 1.65 – 1.85 kg/kg), they exhibited greater ESs for vertical jump height improvements (1.13 vs 0.31) and reductions in COD times (-1.28 vs -0.82) following CNT. Nonetheless, statistical differences were not detected for "athlete calibre" in maximal strength (1-RM), vertical jump height, sprint time, or COD tests for the DT paradigm. Based upon these data, it is likely that CNT is beneficial for both trained/highly trained and recreationally trained team-sport athletes even though it is unclear whether pre-training strength or study design characteristics mediate these adaptations. In contrast, Lesinski et al. [22] found recreationally trained athletes improved CMJ height more than trained/highly trained athletes ($+9.7\%$ vs $+2.7\%$) after CNT. Neither of these studies [22, 26] appropriately assessed whether the athletes that were deemed to be trained/highly trained had the requisite levels of strength required to maximise the benefits of these training methods (controlled for strength). Thus, it is possible that the athletes' strength levels, rather than performance calibre per se, was the mediating factor underpinning the greater adaptation to the CNT.

These benefits may be associated with their muscle fibre type composition whereby athletes with a greater strength levels (and cross sectional area and volume), who may most

likely be at the higher end of performance calibre classification (e.g., elite), may have a greater percentage of skeletal muscle type II (fast-twitch) myosin isoform ($\uparrow$ MRLC phosphorylation, blood flow, osmolarity, and temperature rise) [99], hence enabling them to rapidly produce greater force in each set, thus potentially conferring greater adaptations [30]. An accumulation of these PAPE-enhanced sessions, in which greater mechanical force and power are expressed, might theoretically explain the superior chronic adaptations in this population. Furthermore, trained athletes may also be more familiarised with the exercises used in training than recreationally trained amateur athletes due to their accumulated training experience. Accordingly, they may be able to perform more work to fatigue following conditioning contractions and thus more frequently use CAs before low-load, higher-velocity exercise sets. Furthermore, well trained individuals may be able to perform more CAs and subsequent sets with higher intensity and might utilise CAs in more sets because they show less fatigue and can handle the greater loads. In summary, practitioners should consider the athlete's performance calibre when considering which form of CPX should be programmed; however, the ideal cohort of athletes for implementation of CNT, is one where the athletes possess the requisite levels of strength to mitigate to inhibitory role of fatigue in potentiation.

4.5 Movement Pattern

It is commonly thought that when using CNT, the CA must be biomechanically similar to the subsequent exercise intended for potentiation (e.g., half-squat paired with vertical CMJ). The reasoning behind this is often explained with terminology commonly used to describe exercises with similar movement patterns—goals or objectives (e.g., vertically oriented vs horizontally oriented) [100]. For instance, if an athlete completes vertically oriented exercises (e.g., back squat), the similarities in the specificity of force application would result in greater potentiation (or training effects) in vertically oriented activities (e.g., vertical jump). Accordingly, it would be expected that exercises with emphasis on antero-posterior force application (e.g., broad jump) would lead to greater acute or chronic improvements in activities in which the hip extensor muscles play a crucial role, such as standing long jumps or short linear sprints. One might expect reasonable similarity in the muscles used, muscle length ranges required, and synergist muscle coordination. However, potentiation is not always present when employing biomechanically similar exercises [101]. Additionally, contraction mode specificity may speculatively play a role, given that exercises performed using one contraction mode may interfere with one using a different contraction mode that is performed soon after [63]. Any motor pattern interference resulting from exercise pairs utilising different movement

patterns (or having different goals) or contraction modes may underpin the lag in performance enhancement usually observed following a CA, or reduce the overall magnitude of performance enhancement. Along the same line, movement velocity could potentially result in motor pattern interference in contrast pairings if the exercise velocities and motor pattern are dissimilar and short recovery periods present. Conversely, if the initial movement is similar but subtly different to the subsequent task, it is conceivable that a motor pattern interference effect can also be present (e.g., cycling before running) [102, 103], and given the great muscle activations in strength training movements, it is feasible that this effect could be amplified. However, the authors acknowledge that additional research in this area is warranted to fully corroborate such a suggestion. Despite the previous, in many cases, incorporating contrast pairings into a strength training programme does not reduce central nervous system excitability nor performance post-CA [79, 104], thus providing a useful strategy for incorporating strength and plyometric exercises within a session. Based on the current evidence, it is rarely possible to accurately predict the potentiating effect of a task with a movement pattern characteristic on a task with another set of characteristics, although greater similarity appears to be beneficial.

4.6 Time Efficiency

Several practical time management strategies can be recommended based on the available scientific evidence. In the S&C community, mixed-training approaches such as CPX, are widely used [105, 106]. Recent surveys of the evidence-based practices of experienced S&C professionals show that CPX methods contribute a large proportion of the training conducted to improve power and speed qualities in professional sports [2]. Interestingly, in some countries, a larger percentage of S&C professionals prefer CNT over DT and AT (e.g., 49%, 31%, and 27%, respectively, based on a survey in Brazilian elite soccer), to incorporate plyometrics with heavier-load strength exercises [107]. Similarly, in an online survey including S&C coaches from professional soccer leagues in 18 countries, CPX was used to programme plyometrics into their workout programmes [105]. CNT is also a commonly used method for sprint development in athletes in elite rugby football codes (prescribed with intention to elicit PAPE) [106]. These studies also highlighted that the coaches programmed CPX strategies specifically to improve “time-efficiency” in periods of the season where the demands of training and congestion of fixtures may reduce the time available to conduct S&C session, thus enabling them to work on various force–velocity characteristics simultaneously.

Furthermore, when working with large groups of athletes, a time-efficient strategy when using contrast loading is to

include a biomechanically dissimilar exercise between the CA and the subsequent low-load, higher-velocity exercises targeted to elicit PAPE where there would alternatively be a complete rest period [108]. As previously mentioned, long rest intervals (> 5 min) are often imposed to realise the PAPE phenomenon. However, it is not practical to impose such rest periods when working with large groups or when athletes are time poor, and therefore, inserting an exercise that targets different musculature may be a viable suggestion for practitioners. For example, when aiming to potentiate the lower body function, either upper body strength (e.g., push-ups and bench press) or other rehabilitation or prehabilitation exercises (e.g., rotator cuff and banded exercises) could be performed during the rest periods [108]. Although this may be feasible, to our knowledge there is no evidence regarding the impact of performing a biomechanically dissimilar exercise in the intra-contrast rest period. Still, as previously stated, practitioners are encouraged to experiment with their PAPE protocols before implementing them into their CPX programme. This may, conceivably, allow developing more time-efficient (and tailor-made) strategies based on both individual characteristics and group context.

4.7 Programming

As described in the historical perspective, the influence of CPX has its origination in early works by Verkhoshansky and colleagues. Furthermore, the purpose of this training strategy is only realised when the training is effectively organised (i.e., periodisation) [75, 109]. Accordingly, an interesting CPX strategy to manipulate loads within a periodisation scheme is the “conjugate method” (introduced by Verkhoshansky) in which several physical qualities, such as strength and power, are combined [109–111]. Whilst implementing the conjugate method, several physical qualities can be developed in combination in each microcycle; however, the intensity is maximised in only one of those qualities in each mesocycle whilst the others receive maintenance loads [110]. This training model, gives priority to specialised training blocks where specific physical adaptation responses related to sport-specific motor output enhancements (i.e., sport form) are targeted [112]. This is a viable model for S&C professionals wishing to maintain or develop physical qualities in various components of the force–velocity relationship throughout a yearly training plan whilst incorporating CPX.

Indeed, CPX programmes may be suitable to utilise in sports that have long in-season periods (e.g., team-sports), with recent meta-analyses showing that various exercise sequences can be optimal depending on objectives of the training [26, 27]. In many cases, the increase in training frequency and match fixture density may lead to elevated stress levels imposed on the athletes, which leads to a reduction

in the frequency of strength-power sessions depending on the sport context [36, 113]. Thus, a potent training stimulus, such as that provided by CNT specifically, may be an interesting option in periods with reduced resistance training volume and large increases in tactical/technical training load [114]. There appears to be meaningful improvements in COD, jumping, and sprint ability in athletic subjects following low-volume (2 sessions per week) in-season CNT [115] and improvement or maintenance of developed maximal strength [36] whilst some investigations failed to detect such improvements [35]. Nonetheless, it is difficult to draw comparisons between studies as many do not adequately report the additional training done by the athletes or their physical or fatigue state.

Complex training can also be incorporated into rehabilitation, as the inherent load manipulation (targeting specific force or velocity components of the force–velocity relationship) allows practitioners to adjust many variables that can aid in return to training and performance (e.g., inclusion of plyometrics in terminal phase of rehabilitation) [116, 117]. Finally, and most importantly when considering how to implement these methods into a training plan, the purpose of the training is paramount rather than focusing solely on the groupings of exercises, as each will have its own adaptive response and the underpinning rationale of implementation may differ depending on contextual factors.

5 Directions for Future Research

Although many acute and longitudinal studies can be found in the sports science literature, further work is needed to determine the effectiveness of these CPX methodologies. First, using the proposed terminology, which is based on the historical perspective of relevant literature, researchers and practitioners can be consistent when sharing knowledge and findings. Second, there are many shortcomings in the current literature that should be considered when interpreting results. Importantly, in acute studies aiming to investigate PAP or PAPE phenomena, the methodological design characteristics (e.g., warm-up, rest intervals, validity of verification of potentiation) must be considered to avoid misinterpretation of the true mechanism occurring in the recovery time between the CA and the performance of the subsequent task [29]. While there is an abundance of CPX research, it is difficult to identify the effectiveness and applicability of this training method with many shortcomings arising from a lack of randomised control trials, no assessor blinding, and possible selective reporting of outcomes [26, 48]. Moreover, it is difficult to make inferences on the effectiveness of any of these methods when summarising the findings (in meta-analyses, for example) and establish a dose–response relationship for these training methods since

inconsistent terminology is used in the calibre and training status of the athletes. As such, uniform language should also be used in this regard [57]. When considering performance calibre and training status, there is an abundance of research on recreationally active to trained subjects but a considerably lower number of investigations on elite and world-class populations, which would offer interesting insights on the effectiveness of this type of programming in athletes with (theoretically) a higher training age in both S&C and competitive sport. Methodological considerations also impact interpretation of the long-term adaptations from such sequences. Furthermore, there is a paucity of CPX research in female participants [118]; thus, future research is needed given that training responses may differ, although adequate study design with analyses addressing the impact of participant characteristics such as training age or skill level and measures of initial strength should also be incorporated to minimise the risk of drawing incorrect conclusions from the data acquired in female participants.

An important methodological change that might improve our understanding of the effects of CPX may be to include the quantification of external (i.e., physical work) and/or internal (i.e., psychophysiological stress response) training loads from sport-specific training and competition [119], thus controlling for these variables when exploring the effectiveness of a CPX intervention. The actual training loads prescribed in the intervention study should be quantified and reported to determine causal relationships between the training performed and the intervention outcomes [120]. There are methods of quantifying internal training loads without the use of expensive laboratory equipment. Session ratings of perceived exertion ($sRPE = RPE \times \text{session duration}$) may be useful [121]. An example of such methodology is represented by Spinetti et al. [24], who compared CNT to traditional strength training and reported their findings with sRPE from both experimental groups (controlling for sport-specific training). Notably, in this instance no between-group improvements reached statistical significance in 5, 10, 20, 30 m sprints, COD speed, or SJ performance. Thus, in the previous CPX example and others [35], accounting for training load may have assisted in the avoidance of incorrect interpretation and inferences regarding these training strategies. Future research should report training loads so that comparisons can be made across different studies whilst considering the demands from the tactical sessions, technical sessions, and matches [122].

Another avenue worth exploring is the use of time-efficient CNT strategies such as inserting a biomechanically dissimilar exercise within the intra-contrast rest period. Currently, no research has documented the impact of such insertion (i.e., motor pattern interference or excessive fatigue hindering the potential PAPE mechanism) when the intended adaptation to the training is to improve muscular power

output. Whereas, if the adaptation is to improve strength and hypertrophy, there are many outlined recommendations on ways to manipulate variables within the training programme design to improve time efficiency (i.e., drop-sets, supersets, etc.) [123]. Furthermore, when contemplating AT or DT, no consensus exists on optimal methods to prescribe the exercises using this strategy. The main question that arises in relation to DT is whether the heavier-load exercises might negatively impact the lighter, high-speed exercises performed at the end of the session, or vice versa. It is often recommended that complex exercises requiring a high skill level (e.g., weightlifting and its derivatives) be performed before fatigue accumulates since fatigue may interfere with movement pattern execution and thus reduce training effectiveness and long-term adaptation [124]. With AT, it is unclear whether performing lighter-load, higher-speed exercises early in the session promote a “warm-up” effect or even if this type of sequence could favour neuromuscular adaptations in high-velocity function since there is minimal fatigue present at beginning of the workout. Several investigations hint at possible mechanisms underpinning this effect [38, 59]; however, no concrete conclusions can be made at present.

6 Conclusion

Training strategies incorporating heavy and light-to-moderate loads with various set structures (such as CPX) have become popular methods in terms of scientific investigations with sport science researchers and S&C professionals. However, as indicated previously, there are disparities in the terminology to define and describe the sequences of these training types. Thus, the following terminology is proposed, based on the genesis analysis and logical reasoning. Namely, we propose:

- Complex training (CPX): umbrella term with 4 different implementations, generally used to indicate a method in which movement velocity or load is altered between sets and/or exercises within the same session with the aim of improving slow and fast force expression;
- Contrast training (CNT): specific type of CPX that consists of alternating high-load and low-load (higher-velocity) exercises in a set-by-set fashion within the same session;
- Ascending training (AT): subset of CPX in which several sets of low-load, higher-velocity exercises are completed before several sets of high-load exercises within the same session;
- Descending training (DT): type of CPX consisting of several sets of high-load (e.g., back squat) exercises completed before the execution of several sets of low-load,

higher-velocity (e.g., vertical jump) exercises within the same session;

- French contrast training (FCNT): subset of CPX-CNT in which a series of exercises is performed in the sequence in the same session: heavy compound exercise, SSC plyometric exercise, light-to-moderate load compound exercise that maximises movement speed (i.e., external power), and a plyometric exercise (often assisted).

Uniform terminology will aid researchers when communicating their findings as well as dissemination to practitioners in the field. It is also important to consider that many aspects of these training methods can be manipulated based on the athlete's performance calibre/strength, and sport, therefore the adaptations might be different depending on the team and individual's needs. Ascending training and DT are recommended to develop appropriate levels of strength and training age before advancing to more advanced training strategies such as CNT (or FCNT) training. Specifically, with initial strength levels (relative strength ~ 2.0 kg/kg for the back squat and ≥ 1.35 kg/kg for bench press with 5–7 min intra-contrast rest) and with adequate resistance training experience. Although the time-course to peak performance enhancement aligns with the mentioned recommendation, with stronger athletes, enhancements in shorter intra-contrast rest intervals (~ 1 –5 min) may still be present and relevant for longer-term adaptations (i.e., effective, and time-efficient). Therefore, practitioners are encouraged to experiment to find the optimal intra-contrast rest period for their athletes before implementing an intervention. In weaker individuals, ≥ 8 min intra-contrast rest is recommended, which may not be practical. Also, modifications can be made to CNT to improve time efficiency within a session; however, caution must be taken as there may be an important motor pattern interference or PAPE inhibition. Also, a high magnitude of inter- and intra-individual variability in the potentiation and the long-term adaptations from these types of training is present and should be considered in future research. Furthermore, there is an abundance of evidence on the effectiveness of these methods; thus, practitioners should use the available literature to guide their training practices whilst keeping in mind their underpinning mechanisms. Going forward, researchers and practitioners should consider using the terminology outlined herein for uniformity.

Declarations

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